

Course Code	Course Title							
116h54C701	Advanced Machine Learning							
	TH			P	TUT			Total
Teaching Scheme(Hrs.)	03			2	--			05
Credits Assigned	03			1	--			04
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	--	--	--	--	100

Course prerequisites: Probability concepts, calculus, linear algebra & basic python programming.

Course Objectives:

6. To analyse various machine learning algorithms and techniques with a modern outlook focusing on recent advances.
7. Explore supervised and unsupervised learning paradigms of machine learning.
8. To understand natural language processing and to learn how to apply basic algorithms in this field.
9. To develop the basic skills necessary to pursue research in machine learning.

Course Outcomes

On completion of the course students will be expected to

1. Describe the fundamental issues and challenges of machine learning: data, model selection, model complexity.
2. Understand & describe mathematical foundation behind deep learning architectures.
3. Design & implement various deep supervised learning architectures for image data.
4. Learn and train various deep learning models in NLP.
5. Apply various deep learning techniques to design efficient algorithms for real-world applications.

Module No.	Unit No.	Details	Hrs.	CO
1	Machine learning foundation:		05	CO1
	What is Machine Learning? Types of learning, applications, Bias, variance, overfitting, underfitting, cross validation and feature engineering, gradient descent learning algorithm and its variations			
2	Deep learning fundamentals:		10	CO2
	Mathematical foundations of deep learning & neural network playground. Learning rates and data normalization, activation functions, Optimizers, Regularization, Dropout, Momentum & Batch Norm. Neural Nets-Deep Vs Shallow Networks- deep multi-layer perceptron. #Self-Learning: forward and backward propagation in neural networks.			
3	Deep Learning Architecture:		10	CO3
	Convolutional Neural Networks: Convolution, pooling operations, basic CNN architecture and image classification using CNN. Recurrent Neural Networks: RNN introduction, difference between feed forward and RNN, forward, backward propagation in RNN & Vanishing/Exploding gradient problem. Long Short Term Memory: LSTM gates, forward, backward propagation in LSTM. #Self-Learning: Auto encoders			
4	Deep learning for NLP:		10	CO4
	NLP introduction, text preprocessing techniques, text representation (one hot encoding, bag of N grams, TFIDF features). Word Embedding – word2vec model. Text generation, Seq2Seq models (Encoder decoder architecture). #Self-Learning: NLTK & Gensim library			
5	Machine learning case studies:		10	CO5
	- Churn Analysis and Prediction - Sentiment Analysis or Topic Mining from New York Times - Customer Segmentation and Value - Netflix Movie Recommendation System - Image & text classification using TensorFlow & PyTorch. - Transfer learning with Pre-trained CNN using TensorFlow.			
Total			45	

Recommended Books:

Sr.	Name/s of Author/s	Title of Book	Name of Publisher	Edition and
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Somaiya Vidyavihar University
K. J. Somaiya College of Engineering, Mumbai -77
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No.			with country	Year of Publication
1	Tom M.Mitchell	Machine Learning	McGraw Hill	2017
2	M. Gopal	Applied Machine Learning	McGraw Hill	2018
3	Ian Goodfellow, Yoshua Bengio, Aaron Courville	Deep Learning	An MIT Press book	2016
4	Deng & Yu	Deep Learning: Methods and Applications	Now Publishers	2013
5	Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta and Harshit Surana	Practical Natural Language Processing	O'Reilly Media, Inc.	June 2020
6	Sudharsan Ravichandiran	Hands-On Deep Learning Algorithms with Python	Packt Publishing	July 2019